

KVault-DB

A Comprehensive, Step-by-Step Implementation Guide
to a Full-Stack C++20 LSM-Tree Key-Value Store

*From shared-memory primitives to a live React dashboard —
every layer, every line, explained.*

Systems Engineering Documentation
Based on the KVault-DB open-source codebase

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Abstract

KVault-DB is a full-stack, single-node Key-Value Store built from first principles in modern C++20. Its storage engine implements the Log-Structured Merge-Tree (LSM-Tree) architecture — the same foundational design used in production systems such as LevelDB, RocksDB, and Apache Cassandra. The system exposes its engine through a multi-threaded HTTP REST API (built with the Crow microframework) and pairs it with a real-time React/Vite visualization dashboard.

This document dissects the implementation of KVault-DB into ten logical, sequential construction steps. Each step covers the theoretical motivation for the design choice, the practical build process, the key architectural decisions, and the exact C++ code extracted directly from the source files. By the end, the reader will have a complete, bottom-up understanding of how a production-grade storage engine is assembled from raw system primitives.

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1 Step 1: Foundation — Core Types and Engine Configuration

1.1 Conceptual Overview

Before writing a single line of business logic, a storage engine must establish a shared vocabulary. Every layer of the system — from the WAL serializer to the HTTP API — must agree on what a *record* is and what types of mutations are possible. Defining these types centrally prevents circular dependencies and eliminates the risk of subtle semantic mismatches between layers.

The most important design choice at this layer is the **Tombstone Sentinel**. In an LSM-Tree, data is *never physically deleted* at insertion time. The storage is append-only. A `DELETE("x")` operation is not a removal; it is the *insertion of a deletion marker* (a tombstone). The system then handles the semantics of deletion during reads and background compaction.

1.2 The Building Process

The foundation consists of two files:

- `include/kvault/types.hpp` — primitive type aliases and the `KVRecord` struct.
- `include/kvault/config.hpp` — the `EngineConfig` struct with all tuneable parameters.

The `EngineConfig` struct acts as a dependency-injection container. It is constructed in `main.cpp` and passed into the `KVStore` constructor, which distributes the relevant fields to each subsystem (WAL, MemTable, SSTable Manager). This design means no subsystem reads global variables, making the entire engine independently testable.

1.3 Implementation Details

The `RecordType` enum uses `uint8_t` as its underlying type. This is not an accident: the WAL binary serializer writes exactly one byte for the record type field. Using `uint8_t` guarantees that the C++ type and the binary wire format are identical, eliminating padding or alignment surprises.

The tombstone sentinel `kTombstoneValue` is a `constexpr string_view` with sentinel bytes (`0x7F`) surrounding a unique marker string. It is stored as the *value* of a deleted key inside the SkipList. This design avoids a separate boolean field on each node, keeping the node layout flat. When the MemTable reads back this sentinel, it interprets the key as deleted and returns `std::nullopt` to the caller.

1.4 Code & Configuration

```

1 #pragma once
2 #include <stdint>
3 #include <string>
4 #include <string_view>
5
6 namespace kvault {
7
8 // Both are variable-length strings. Keys are compared
9 // lexicographically
10 // for ordered storage in the Skip List and SSTables.
11 using Key    = std::string;
12 using Value  = std::string;
13
14 // Stored as a single byte in the WAL binary format and SSTable records
15 // .
16 // uint8_t ensures exact 1-byte serialization with no padding.

```

```

15 enum class RecordType : uint8_t {
16     PUT      = 0x00, // Insert or overwrite a key-value pair
17     DELETE   = 0x01, // Remove a key (written as a tombstone in the
        MemTable)
18 };
19
20 // The universal mutation unit. Every write flowing through the system
21 // is represented as a KVRecord.
22 struct KVRecord {
23     RecordType type;
24     Key         key;
25     Value       value; // Empty for DELETE records
26 };
27
28 // Used internally by the MemTable to mark deleted keys.
29 // 0x7F sentinel bytes make collision with real user data virtually
        impossible.
30 // NEVER exposed to external callers.
31 inline constexpr std::string_view kTombstoneValue =
32     "\x7F__KVAULT_TOMBSTONE__\x7F";
33
34 } // namespace kvault

```

Listing 1: Core Types: include/kvault/types.hpp

```

1 namespace kvault {
2
3 struct EngineConfig {
4     // Approximate byte threshold at which the MemTable is flushed to
        disk.
5     // Default: 4 MB. Larger values increase write throughput (fewer
        flushes)
6     // but consume more RAM.
7     size_t memtable_flush_threshold_bytes = 4UL * 1024 * 1024;
8
9     // Maximum Skip List levels. 16 levels supports up to ~65K entries
        // per MemTable flush cycle.
10    int skip_list_max_level = 16;
11
12    // WAL and SSTable storage directories.
13    std::string wal_directory      = "data/wal";
14    std::string sstable_directory = "data/sstables";
15
16    // If true, every WAL append issues fsync() to guarantee stable
        storage.
17    // If false, faster but risks losing last few records on power
        failure.
18    bool sync_per_write = true;
19
20    // Bits allocated per key in each Bloom Filter.
21    // 10 bits/key -> ~0.8% false positive rate (k ~= 7 hash functions)
        .
22    size_t bloom_filter_bits_per_key = 10;
23
24    // Port for the embedded Crow HTTP server.
25    uint16_t server_port = 8080;
26 };
27
28

```

```
29 } // namespace kvault
```

Listing 2: Engine Configuration: `include/kvault/config.hpp` (excerpt)

2 Step 2: The Probabilistic In-Memory Index — The Skip List

2.1 Conceptual Overview

The MemTable needs a data structure that keeps keys *sorted* at all times (so flushing to a sorted SSTable is a simple sequential scan) while supporting $O(\log n)$ insertions, deletions, and exact-key searches.

A **Skip List** is a probabilistic ordered data structure invented by William Pugh (1990). It is a layered linked list where each node participates in one or more *express lanes* at higher levels. A search starts at the highest level, skipping large chunks of the list, then descends to lower levels as it narrows in on the target key. The expected complexity for all operations is $O(\log n)$, equivalent to a balanced binary search tree but with a far simpler implementation that is amenable to concurrent access.

A node of height h is promoted to level $h + 1$ with probability $p = 0.5$. This gives an expected node height of $\frac{1}{1-p} = 2$, meaning the expected total number of forward pointers across all n nodes is $2n$ — highly space-efficient.

2.2 The Building Process

A naïve implementation allocates each node with `new` and deletes them recursively. For large lists (millions of nodes), the recursive destructor causes a **stack overflow**. KVault solves this with an **Arena Allocator**: every node is stored in a `std::vector<std::unique_ptr<SkipListNode>>` called `arena_`. When the list is destroyed, the vector’s destructor calls `delete` on each node iteratively, not recursively.

2.3 Implementation Details

The critical insight is the **ownership split**:

- `arena_` holds `unique_ptr<Node>` — it *owns* every node.
- `forward[]` inside each node holds raw `Node*` pointers — non-owning, for traversal only.

Using `unique_ptr` for `forward[]` would be incorrect because the same node is reachable from multiple levels simultaneously. Two `unique_ptr`s pointing to the same node would cause a double-free Undefined Behaviour.

The `insert()` algorithm works by building an `update[]` vector recording the last node at each level whose key is less than the target. After the traversal, a new node is spliced between `update[i]` and `update[i]->forward[i]` at each level.

2.4 Code & Configuration

```
1 struct SkipListNode {
2     Key    key;
3     Value  value;
4
5     // Non-owning traversal pointers. Lifetime managed by SkipList::
6     // arena_.
7     // Using raw pointers here is correct: the same node can be pointed
8     // to
9     // from multiple levels, so unique_ptr would cause a double-free.
```

```

8      std::vector<SkipListNode*> forward;
9
10     // Data node constructor
11     SkipListNode(Key k, Value v, int height)
12         : key(std::move(k))
13         , value(std::move(v))
14         , forward(static_cast<size_t>(height), nullptr) {}
15
16     // Sentinel (head) node constructor
17     explicit SkipListNode(int height)
18         : forward(static_cast<size_t>(height), nullptr) {}
19 };
20
21 class SkipList {
22 public:
23     static constexpr int      kMaxLevel      = 16;
24     static constexpr double   kProbability   = 0.5;
25
26     // Destructor: arena_ iteratively destroys all nodes (O(n), stack-
27     safe).
28     ~SkipList() = default;
29
30     void          insert(const Key& key, const Value& value);
31     std::optional<Value> search(const Key& key) const;
32     bool          remove(const Key& key);
33 private:
34     int random_level();
35
36     std::unique_ptr<SkipListNode> head_; // Sentinel, entry point at
37     all levels
38     std::vector<std::unique_ptr<SkipListNode>> arena_; // OWNS all data
39     nodes
40     int      current_level_;
41     size_t   size_;
42     mutable std::mt19937 rng_;
43     mutable std::uniform_real_distribution<double> dist_;
44 };

```

Listing 3: Skip List Node and Class Declaration: include/kvault/skiplist.hpp

```

1 int SkipList::random_level() {
2     int level = 1;
3     while (level < kMaxLevel && dist_(rng_) < kProbability)
4         ++level;
5     return level;
6 }
7
8 void SkipList::insert(const Key& key, const Value& value) {
9     // update[i] = rightmost node at level i whose key < target.
10    std::vector<SkipListNode*> update(kMaxLevel, nullptr);
11    SkipListNode* current = head_.get();
12
13    // Traverse: start at highest level, descend when key would
14    overshoot.
15    for (int i = current_level_ - 1; i >= 0; --i) {
16        while (current->forward[i] && current->forward[i]->key < key)
17            current = current->forward[i];

```

```

17     update[i] = current;
18 }
19
20 // Upsert: if key exists at level 0, overwrite in place.
21 SkipListNode* candidate = current->forward[0];
22 if (candidate && candidate->key == key) {
23     candidate->value = value;
24     return;
25 }
26
27 // Allocate new node in the arena (unique ownership).
28 const int new_height = random_level();
29 if (new_height > current_level_) {
30     for (int i = current_level_; i < new_height; ++i)
31         update[i] = head_.get();
32     current_level_ = new_height;
33 }
34
35 arena_.push_back(std::make_unique<SkipListNode>(key, value,
36     new_height));
37 SkipListNode* new_node = arena_.back().get(); // raw ptr for wiring
38
39 // Splice new_node between update[i] and update[i]->forward[i].
40 for (int i = 0; i < new_height; ++i) {
41     new_node->forward[i] = update[i]->forward[i];
42     update[i]->forward[i] = new_node;
43 }
44 ++size_;
45 }
46
47 std::optional<Value> SkipList::search(const Key& key) const {
48     const SkipListNode* current = head_.get();
49     for (int i = current_level_ - 1; i >= 0; --i)
50         while (current->forward[i] && current->forward[i]->key < key)
51             current = current->forward[i];
52
53     const SkipListNode* candidate = current->forward[0];
54     if (candidate && candidate->key == key)
55         return candidate->value;
56     return std::nullopt;
57 }

```

Listing 4: Skip List Insert and Search: src/skiplist.cpp

3 Step 3: The In-Memory Write Buffer — The MemTable

3.1 Conceptual Overview

The **MemTable** is the active, mutable tier of the LSM-Tree. Every incoming PUT and DELETE request is directed here first. It acts as a write buffer: mutations accumulate in memory until the table reaches a configurable byte threshold (e.g., 4 MB), at which point it is *frozen* (made immutable) and flushed to disk as a new SSTable file.

This design transforms the write problem from a random-write workload (the bane of mechanical hard drives and flash storage) into a sequential write workload, dramatically improving throughput. The trade-off is read amplification: a lookup may need to search the MemTable and multiple SSTable files on disk.

3.2 The Building Process

The MemTable class wraps the SkipList and adds:

1. **Byte tracking:** it maintains a running `approximate_size_` counter to know when to trigger a flush.
2. **Tombstone logic:** `remove()` inserts the key with `kTombstoneValue` as its value.
3. **Snapshot generation:** `snapshot()` walks the Skip List's sorted level-0 chain and emits a `std::vector<KVRecord>` for the SSTable writer.

3.3 Implementation Details

The tombstone design is central to LSM-Tree correctness. When a user calls `DELETE("x")`, the key might exist in an older SSTable on disk. If the MemTable simply performed a physical removal from the Skip List, the subsequent read path would find nothing in the MemTable and then *fall through to the SSTable and return stale data*. By inserting the tombstone, the MemTable short-circuits this: `contains_tombstone(key)` returns `true`, and the KVStore never initiates the expensive disk search.

3.4 Code & Configuration

```

1 void MemTable::put(const Key& key, const Value& value) {
2     auto existing = skiplist_.search(key);
3     if (existing.has_value()) {
4         // Upsert: node overhead unchanged; only value size delta
5         // matters.
6         approximate_size_ -= existing->size();
7         approximate_size_ += value.size();
8     } else {
9         // New entry: account for key, value, and node pointer overhead
10        approximate_size_ += key.size() + value.size() +
11        kNodeOverheadBytes;
12    }
13    skiplist_.insert(key, value);
14 }
15
16 // Deletes are NOT physical removals. We insert a tombstone sentinel so
17 // the read path can short-circuit SSTable lookups for deleted keys.
18 void MemTable::remove(const Key& key) {
19     auto existing = skiplist_.search(key);
20     if (existing.has_value()) {
21         approximate_size_ -= existing->size();
22         approximate_size_ += kTombstoneValue.size();
23     } else {
24         approximate_size_ += key.size() + kTombstoneValue.size()
25         + kNodeOverheadBytes;
26     }
27     skiplist_.insert(key, std::string(kTombstoneValue));
28 }
29
30 std::optional<Value> MemTable::get(const Key& key) const {
31     auto result = skiplist_.search(key);
32     // A tombstone means the key was deleted; return nullopt to the
33     // caller.
34     if (result.has_value() && *result == kTombstoneValue)
35         return std::nullopt;
36 }

```

```

33     return result;
34 }
35
36 // Used by KVStore to short-circuit expensive SSTable lookups for
37 // deleted keys.
38 bool MemTable::contains_tombstone(const Key& key) const {
39     auto result = skiplist_.search(key);
40     return result.has_value() && *result == kTombstoneValue;
41 }
42
43 // Walk the sorted level-0 chain; produce a sorted KVRecord vector for
44 // the
45 // SSTable writer. This is the 'snapshot' taken before a flush.
46 std::vector<KVRecord> MemTable::snapshot() const {
47     std::vector<KVRecord> records;
48     records.reserve(skiplist_.size());
49     for (auto it = skiplist_.begin(); it != skiplist_.end(); ++it) {
50         KVRecord record;
51         if (it.value() == kTombstoneValue) {
52             record.type = RecordType::DELETE;
53             record.key = it.key();
54         } else {
55             record.type = RecordType::PUT;
56             record.key = it.key();
57             record.value = it.value();
58         }
59         records.push_back(std::move(record));
60     }
61     return records;
62 }

```

Listing 5: MemTable Core Logic: src/memtable.cpp

4 Step 4: Crash Resilience — The Write-Ahead Log

4.1 Conceptual Overview

The MemTable lives in volatile RAM. A server crash — a power failure, an OS panic, or even a `kill -9` signal — would erase all unacknowledged writes. The **Write-Ahead Log (WAL)** is the mechanism that prevents this data loss.

The WAL is an append-only binary file. The *durability contract* is strict:

1. A write request arrives at the KVStore.
2. The mutation is serialized and appended to the WAL file on disk.
3. **Only after** the WAL write is confirmed does the MemTable get updated.
4. The HTTP response is sent to the client.

On startup, the WAL is *replayed*: every record is re-applied to a fresh MemTable, reconstructing the exact in-memory state at the time of the crash.

4.2 The Building Process

The WAL is a self-describing binary log. Each record has a fixed-layout header and a trailing CRC32 checksum. This allows the replay reader to detect *partial writes* — a crash that interrupted the write mid-record leaves a corrupted final record. The replay stops at the first CRC mismatch, safely discarding all records after it.

The CRC32 lookup table is generated at **compile time** using `constexpr`. At runtime, computing a checksum is a simple table lookup with no initialization overhead.

4.3 Implementation Details

The durability is achieved via two levels of flushing:

- `fflush()` — pushes data from the C library’s userspace buffer into the OS kernel’s page cache (fast, in RAM).
- `fsync()` / `_commit()` — forces the OS to write the page cache to the physical storage device (slow, but survives a power failure).

When `sync_per_write = true` (the default), both are called on every append. This limits throughput to the hardware’s IOPS capacity (typically 500–5000 per second for consumer SSDs) but provides 100% crash safety.

4.4 Code & Configuration

```

1 // Standard CRC32, reflected polynomial 0xEDB88320 (same as Ethernet/
  // zlib).
2 // Generated entirely at compile time -- zero runtime initialization
  // cost.
3 constexpr std::array<uint32_t, 256> generate_crc32_table() {
4     std::array<uint32_t, 256> table{};
5     for (uint32_t i = 0; i < 256; ++i) {
6         uint32_t crc = i;
7         for (int j = 0; j < 8; ++j)
8             crc = (crc & 1u) ? (crc >> 1) ^ 0xEDB88320u : (crc >> 1);
9         table[i] = crc;
10    }
11    return table;
12 }
13 constexpr auto kCRC32Table = generate_crc32_table();
14
15 // WAL Binary Record Layout (all multi-byte fields are little-endian):
16 //
17 // Offset  Size  Field          Description
18 // -----  ---  -
19 // 0         1    record_type    0x00 = PUT, 0x01 = DELETE
20 // 1         4    key_length     Length of key_data in bytes (uint32 LE)
21 // 5         K    key_data       Raw key bytes
22 // 5+K       4    value_length   Length of value_data in bytes (uint32 LE)
23 // 9+K       V    value_data     Raw value bytes
24 // 9+K+V     4    crc32         CRC32 over all preceding bytes (uint32
  // LE)
25 //
  // Total: 13 + K + V bytes per record.
26
27 std::vector<uint8_t> WriteAheadLog::serialize_record(const KVRecord&
  record) {
28     const auto key_len = static_cast<uint32_t>(record.key.size());
29     const auto val_len = static_cast<uint32_t>(record.value.size());
30     std::vector<uint8_t> buf;
31     buf.reserve(13 + key_len + val_len);
32
33     buf.push_back(static_cast<uint8_t>(record.type)); // 1 byte type

```

```

34     write_u32_le(buf, key_len);                                // 4 bytes key
35         length
36     buf.insert(buf.end(), record.key.begin(), record.key.end());
37     write_u32_le(buf, val_len);                                // 4 bytes value
38         length
39     buf.insert(buf.end(), record.value.begin(), record.value.end());
40
41     // Append CRC32 of all preceding bytes to detect partial writes.
42     const uint32_t crc = compute_crc32(buf.data(), buf.size());
43     write_u32_le(buf, crc);
44     return buf;
45 }

```

Listing 6: Compile-Time CRC32 Table and WAL Binary Format: src/wal.cpp

```

1  void WriteAheadLog::append(const KVRecord& record) {
2      const auto serialized = serialize_record(record);
3      const size_t written = std::fwrite(
4          serialized.data(), 1, serialized.size(), write_handle_);
5      if (written != serialized.size())
6          throw std::runtime_error("WAL: short write");
7      sync_to_disk();
8  }
9
10 void WriteAheadLog::sync_to_disk() {
11     if (!write_handle_) return;
12     std::fflush(write_handle_);    // Push userspace buffer to OS page
13                                     cache.
14     if (sync_per_write_) {
15 #ifdef _WIN32
16         _commit(_fileno(write_handle_));    // Windows hardware flush
17 #else
18         ::fsync(fileno(write_handle_));    // POSIX hardware flush
19 #endif
20     }
21 }
22
23 // Replay: re-applies all valid WAL records to a fresh MemTable on
24 // startup.
25 std::vector<KVRecord> WriteAheadLog::replay() const {
26     std::vector<KVRecord> records;
27     FILE* read_handle = std::fopen(path_.string().c_str(), "rb");
28     if (!read_handle) return records; // First run, no WAL yet.
29     while (true) {
30         auto record = deserialize_record(read_handle);
31         if (!record.has_value()) break; // EOF or CRC mismatch -- stop
32                                     here.
33         records.push_back(std::move(*record));
34     }
35     std::fclose(read_handle);
36     return records;
37 }

```

Listing 7: WAL Append and Platform Sync: src/wal.cpp

5 Step 5: Read Optimization — The Bloom Filter

5.1 Conceptual Overview

In an LSM-Tree, a GET request that misses the MemTable must search the SSTable files on disk. Disk I/O is orders of magnitude slower than RAM access. A key that has *never been inserted* into the database would cause the system to search every SSTable file before concluding the key does not exist — a needlessly expensive operation.

A **Bloom Filter** is a space-efficient probabilistic data structure that answers the set membership question. Given a set S and a query element x :

- **FALSE** $\Rightarrow x$ is *definitively* not in S .
- **TRUE** $\Rightarrow x$ is *probably* in S (false positives possible).

False negatives are mathematically impossible. This asymmetry makes Bloom Filters a perfect pre-filter for disk I/O: a **FALSE** result eliminates the SSTable search entirely.

With m bits and k hash functions for n keys, the false positive rate is:

$$p = \left(1 - e^{-kn/m}\right)^k$$

Optimal values: $m = -n \ln(p)/(\ln 2)^2$, $k = (m/n) \ln 2$.

With `bits_per_key = 10`: $p \approx 0.0082$ ($\approx 0.8\%$ false positive rate).

5.2 The Building Process

Instead of implementing k independent hash functions (complex and expensive), KVault uses the **Kirsch–Mitzenmacher double hashing technique** [ESA 2006]. Two base hashes h_1 and h_2 are computed from the key using FNV-1a and a Murmur-inspired finalizer. The k composite hashes are then:

$$g_i(\text{key}) = (h_1(\text{key}) + i \cdot h_2(\text{key})) \bmod m, \quad i \in [0, k)$$

5.3 Implementation Details

h_2 is forced to be odd via `h2 |= 1`. This guarantees that h_2 is coprime with any power-of-two m , ensuring the sequence $h_1, h_1 + h_2, h_1 + 2h_2, \dots$ visits m distinct positions before repeating (full coverage of the bit array).

The filter is serialized as [4 bytes: `k`] [4 bytes: `m`] [`ceil(m/8)` bytes: `bits`] and embedded directly into the SSTable file footer region, so it can be loaded into memory without a separate I/O call.

5.4 Code & Configuration

```

1 // Two independent 64-bit hashes from the same key.
2 // h1: FNV-1a -- fast, excellent avalanche for short keys.
3 // h2: Murmur finalizer applied to h1 -- statistically independent.
4 std::pair<uint64_t, uint64_t> BloomFilter::base_hashes(const Key& key)
5 {
6     constexpr uint64_t kFNVPPrime = 0x000001000000001B3ULL;
7     constexpr uint64_t kFNVOffset = 0xCBf29CE48422325ULL;
8
9     uint64_t h1 = kFNVOffset;
10    for (char ch : key) {
11        h1 ^= static_cast<uint64_t>(static_cast<unsigned char>(ch));
12        h1 *= kFNVPPrime;
13    }

```

```

13
14 // Murmur finalizer mix: breaks linear correlations in h1.
15 uint64_t h2 = h1;
16 h2 ^= h2 >> 33; h2 *= 0xFF51AFD7ED558CCDULL;
17 h2 ^= h2 >> 33; h2 *= 0xC4CEB9FE1A85EC53ULL;
18 h2 ^= h2 >> 33;
19 return { h1, h2 };
20 }
21
22 void BloomFilter::add(const Key& key) {
23     auto [h1, h2] = base_hashes(key);
24     h2 |= 1; // Ensure h2 is odd (coprime with m for full bit-array
25              // coverage)
26
27     for (size_t i = 0; i < k_; ++i) {
28         const size_t bit_pos = (h1 + i * h2) % m_;
29         bits_[bit_pos / 8] |= static_cast<uint8_t>(1u << (bit_pos % 8))
30     }
31 }
32
33 bool BloomFilter::might_contain(const Key& key) const {
34     auto [h1, h2] = base_hashes(key);
35     h2 |= 1;
36
37     for (size_t i = 0; i < k_; ++i) {
38         const size_t bit_pos = (h1 + i * h2) % m_;
39         if (!(bits_[bit_pos / 8] & (1u << (bit_pos % 8))))
40             return false; // Definite miss -- key was never add()ed
41     }
42     return true; // Probable hit -- proceed to disk search

```

Listing 8: Bloom Filter Double Hashing: src/bloom_filter.cpp

6 Step 6: Disk Persistence — The SSTable Writer

6.1 Conceptual Overview

When the MemTable reaches its size threshold, it must be written to disk as an immutable **Sorted String Table (SSTable)**. Because the MemTable’s Skip List is already sorted by key, the flush is a simple, single-pass sequential write — one of the most efficient I/O patterns available on any storage medium.

An SSTable is a self-describing binary file. A reader starting from any state can load an SSTable, locate its Bloom Filter, and binary-search for any key without any external metadata.

6.2 The Building Process

The SSTable file layout is written sequentially in four phases:

1. **Data Block:** all key-value records encoded sequentially.
2. **Sparse Index Block:** one index entry per 100 keys (configurable), each storing a key and its byte offset in the data block. Enables $O(\log n)$ block-level lookup.
3. **Bloom Filter Block:** the serialized Bloom Filter (enables $O(1)$ membership test before any data block I/O).

4. **Footer** (fixed 48 bytes): byte offsets of all three blocks, entry count, and a magic number for file integrity validation.

6.3 Implementation Details

The SSTableFooter struct uses `static_assert` to enforce its exact byte size at compile time:

```
static_assert(sizeof(SSTableFooter) == 48)
```

This is a zero-cost compile-time contract. If a developer ever accidentally adds a new field and forgets to update serialization code, the build fails immediately rather than silently producing a corrupted SSTable at runtime.

6.4 Code & Configuration

```

1 // SSTable Binary Layout:
2 //
3 // +-----+ <- byte 0
4 // | DATA BLOCK |
5 // | Sequential KVRecords |
6 // +-----+ <- index_block_offset
7 // | SPARSE INDEX BLOCK |
8 // | [key_len][key][offset] | (every kIndexBlockInterval-th key)
9 // +-----+ <- bloom_block_offset
10 // | BLOOM FILTER BLOCK |
11 // | [k][m][bit_array...] |
12 // +-----+ <- footer_offset
13 // | FOOTER (48 bytes) |
14 // | Block offsets + magic |
15 // +-----+ <- EOF
16
17 class SSTableWriter {
18 public:
19     // One sparse index entry per 100 records. The reader binary-
20     // searches
21     // this index, then linearly scans within the block.
22     static constexpr size_t kIndexBlockInterval = 100;
23
24     // Magic number at end of footer: ASCII "KVL" + version 0x0001.
25     static constexpr uint64_t kMagicNumber = 0x544C564B00010000ULL;
26
27     // Write a sorted vector of KVRecords to a new SSTable file.
28     // sorted_entries MUST be in ascending key order (from MemTable::
29     // snapshot()).
30     static void write(const std::filesystem::path& path,
31                     const std::vector<KVRecord>& sorted_entries,
32                     size_t bits_per_key = 10);
33
34     SSTableWriter() = delete; // Static-only utility class.
35 };
36
37 // Fixed 48-byte footer. static_assert enforces layout at compile time.
38 struct SSTableFooter {
39     uint64_t index_block_offset; // Byte offset of the Sparse Index
40     // Block
41     uint64_t bloom_block_offset; // Byte offset of the Bloom Filter
42     // Block

```



```

39     uint64_t footer_offset;           // Byte offset of this footer itself
40     uint64_t entry_count;             // Total number of records in the
        SSTable
41     uint64_t data_block_size;         // Size of the Data Block in bytes
42     uint64_t magic;                   // kMagicNumber -- validates file
        integrity
43     static constexpr size_t kSerializedSize = 48;
44 };
45 // Compile-time contract: if a field is added, the build fails.
46 static_assert(sizeof(SSTableFooter) == SSTableFooter::kSerializedSize,
47               "SSTableFooter size mismatch -- update kSerializedSize");

```

Listing 9: SSTable File Layout and Footer: include/kvault/sstable_writer.hpp

7 Step 7: Reading from Disk — The SSTable Reader

7.1 Conceptual Overview

Reading from an SSTable is the inverse of writing. The reader must:

1. Load the 48-byte footer to locate the Bloom Filter and Sparse Index.
2. Check the Bloom Filter. If it returns `FALSE`, return immediately.
3. Binary-search the Sparse Index to find the data block offset nearest to the target key.
4. Linearly scan the data block from that offset until the key is found or passed.

This three-tier search strategy minimizes disk I/O. The Bloom Filter lives in RAM. The Sparse Index is small enough to keep in RAM. Only the final linear scan touches the potentially large data block on disk.

7.2 Implementation Details

On construction, the `SSTableReader` reads the footer and loads the Bloom Filter into memory. This is a *one-time cost* paid when the SSTable Manager loads files at startup.

Lexicographic key ordering, which the Skip List and SSTable Writer both preserve, is the key invariant that makes binary search on the Sparse Index correct. In C++, `std::string` comparison is lexicographic by default, so no custom comparator is needed.

7.3 Code & Configuration

```

1  // On construction: load footer + Bloom Filter into RAM (one-time cost)
2  // SSTableReader::SSTableReader(const std::filesystem::path& path) {
3  //     1. Open file.
4  //     2. Seek to EOF - 48 bytes, read footer.
5  //     3. Validate magic number.
6  //     4. Seek to footer.bloom_block_offset, read Bloom Filter bytes.
7  //     5. Deserialize BloomFilter from bytes. Store in bloom_filter_.
8  // }
9
10 // get_record(): The three-tier lookup.
11 // std::optional<KVRecord> SSTableReader::get_record(const Key& key)
    const {
12 //
13 //     // Tier 1: Bloom Filter (in RAM, O(k) time, no disk I/O)
14 //     if (!bloom_filter_.might_contain(key))

```



```

15 //         return std::nullopt; // Definite miss. Zero disk I/O needed.
16 //
17 // // Tier 2: Binary search the Sparse Index (small, usually cached
18 //   by OS)
19 // // to find the data block offset nearest to 'key'.
20 //   uint64_t scan_offset = binary_search_index(key);
21 //
22 // // Tier 3: Linear scan the data block from scan_offset.
23 // // Decode records sequentially until key matches or key is passed.
24 //   seek_to(scan_offset);
25 //   while (not_at_index_block) {
26 //       KVRecord rec = decode_next_record();
27 //       if (rec.key == key) return rec; // Found
28 //       if (rec.key > key) return std::nullopt; // Passed it -- not
29 //         present
30 //   }
31 //   return std::nullopt;
32 // }

```

Listing 10: SSTable Search Strategy (conceptual from `src/sstable_reader.cpp`)

8 Step 8: Managing Generations — The SSTable Manager

8.1 Conceptual Overview

As the engine runs, each MemTable flush produces a new `.sst` file on disk. Over time, there may be dozens or hundreds of them. The **SSTable Manager** is responsible for keeping track of all active SSTable files and executing the *cascading read path*: searching SSTables from newest to oldest until the key is found.

The generation ordering is critical. If key "x" was written with value "v1" and later overwritten with "v2", these may reside in different SSTables. Searching newest-to-oldest ensures that "v2" (the newer, correct value) is returned without ever reading the stale "v1" record.

8.2 Implementation Details

SSTable filenames are zero-padded millisecond timestamps (e.g., `0001691234567000.sst`). This ensures that lexicographic filename ordering matches chronological ordering. On startup, the Manager sorts files in *descending* lexicographic order (newest first) so that `readers_[0]` is always the most recent SSTable.

When a new SSTable is flushed, it is prepended to `readers_` at index 0 — maintaining the newest-first invariant without re-sorting.

8.3 Code & Configuration

```

1 SSTableManager::SSTableManager(const std::filesystem::path& sstable_dir
2   )
3   : sstable_dir_(sstable_dir)
4   {
5       std::filesystem::create_directories(sstable_dir_);
6
7       // Collect all .sst files in the directory.
8       std::vector<std::filesystem::path> sst_files;
9       for (const auto& entry : std::filesystem::directory_iterator(
10         sstable_dir_)) {

```

```

9         if (entry.is_regular_file() && entry.path().extension() == ".
            sst")
10             sst_files.push_back(entry.path());
11     }
12
13     // Sort descending (newest first) by lexicographic filename.
14     // Zero-padded timestamps ensure this is also chronological order.
15     std::sort(sst_files.begin(), sst_files.end(), std::greater<>());
16
17     for (const auto& path : sst_files)
18         readers_.push_back(std::make_unique<SSTableReader>(path));
19 }
20
21 // Cascading read: search newest to oldest.
22 // The first match wins -- it represents the most recent state of the
    key.
23 std::optional<KVRecord> SSTableManager::get_record(const Key& key)
    const {
24     for (const auto& reader : readers_) {
25         auto result = reader->get_record(key);
26         if (result) {
27             // Found! Could be PUT or DELETE (tombstone from an older
                flush).
28             return result;
29         }
30     }
31     return std::nullopt;
32 }
33
34 // A newly flushed SSTable is always the newest generation.
35 // Prepend to readers_ to maintain newest-first ordering.
36 void SSTableManager::add_sstable(const std::filesystem::path& path) {
37     auto reader = std::make_unique<SSTableReader>(path);
38     readers_.insert(readers_.begin(), std::move(reader));
39 }

```

Listing 11: SSTable Manager: Startup Scan and Cascading Read: src/sstable_manager.cpp

9 Step 9: The Concurrent Engine Core — KVStore

9.1 Conceptual Overview

The **KVStore** class is the central orchestrator of the entire storage engine. It owns and coordinates the MemTable, the WAL, and the SSTable Manager behind a single, thread-safe public interface. Because the Crow HTTP server handles many client requests concurrently, the **KVStore** must be able to safely serve multiple readers and writers simultaneously.

The concurrency strategy is a classic **Reader-Writer lock** (`std::shared_mutex`):

- **Readers** (`get`) acquire a `std::shared_lock` — multiple readers can proceed simultaneously.
- **Writers** (`put`, `remove`, `flush`) acquire a `std::unique_lock` — exclusive access, blocking all readers and other writers.

9.2 Implementation Details

The `remove()` method makes a critically correct design decision. In a naive implementation, one might first call `get()` to check if the key exists before deleting it. This would deadlock:

`get()` acquires a `shared_lock`, but `remove()` already holds a `unique_lock`, and a single thread cannot hold both simultaneously in `std::shared_mutex`.

The solution is a *blind write*: the tombstone is written unconditionally, regardless of whether the key exists. This is semantically correct in an LSM-Tree (a DELETE in an empty set is a no-op on replay) and eliminates the deadlock. This is the same approach used by all production LSM-Tree databases.

The flush pipeline (`trigger_flush()`) executes under the `unique_lock` held by the calling write operation. It snapshots the MemTable, writes the SSTable to disk, registers the new file with the Manager, clears the MemTable, and truncates the WAL.

9.3 Code & Configuration

```

1 void KVStore::put(const Key& key, const Value& value) {
2     std::unique_lock lock(rw_mutex_); // Exclusive: blocks all readers/
        writers
3
4     // WAL first -- data is durable on disk before MemTable is updated.
5     wal_>append(KVRecord{RecordType::PUT, key, value});
6     memtable_>put(key, value);
7
8     if (memtable_>should_flush() ||
9         wal_>file_size() >= config_.memtable_flush_threshold_bytes)
10        trigger_flush();
11 }
12
13 std::optional<Value> KVStore::get(const Key& key) const {
14     std::shared_lock lock(rw_mutex_); // Shared: multiple readers
        allowed
15
16     // Step 1: Tombstone short-circuit -- avoids expensive disk search.
17     if (memtable_>contains_tombstone(key))
18         return std::nullopt;
19
20     // Step 2: Search active MemTable.
21     auto mem_val = memtable_>get(key);
22     if (mem_val) return mem_val;
23
24     // Step 3: Fall back to SSTable Manager (newest-to-oldest disk
        search).
25     auto sst_val = sstable_manager_>get(key);
26     if (sst_val && sst_val->empty()) return std::nullopt; // SSTable
        tombstone
27     return sst_val;
28 }
29
30 bool KVStore::remove(const Key& key) {
31     std::unique_lock lock(rw_mutex_);
32
33     // BLIND WRITE: write tombstone unconditionally.
34     // Do NOT call get() here -- that would require shared_lock,
        causing
35     // a deadlock since we already hold unique_lock.
36     wal_>append(KVRecord{RecordType::DELETE, key, ""});
37     memtable_>remove(key);
38
39     if (memtable_>should_flush() ||

```

```

40     wal_>file_size() >= config_.memtable_flush_threshold_bytes)
41     trigger_flush();
42     return true;
43 }
44
45 // Flush pipeline: snapshot -> write SSTable -> register -> reset.
46 void KVStore::trigger_flush() {
47     // Caller MUST hold unique_lock on rw_mutex_.
48     auto records = memtable_>snapshot();
49     if (records.empty()) return;
50
51     auto sst_path = std::filesystem::path(config_.sstable_directory)
52                   / generate_sstable_filename();
53     SSTableWriter::write(sst_path, records, config_.
54                          bloom_filter_bits_per_key);
55     sstable_manager_>add_sstable(sst_path);
56
57     // Reset MemTable and truncate WAL (records now safely in SSTable).
58     memtable_ = std::make_unique<MemTable>(config_.
59        memtable_flush_threshold_bytes);
60     wal_>truncate();
61 }

```

Listing 12: Thread-Safe Write and Read Paths: `src/kvstore.cpp`

10 Step 10: Full-Stack Integration — REST API and React Dashboard

10.1 Conceptual Overview

The final step connects the C++ storage engine to the outside world via a standard HTTP REST API and pairs it with a real-time React visualization dashboard. The API transforms KVault from an embedded library into a *network database service*: any application in any language can store and retrieve data using standard HTTP requests.

The Crow C++ microframework handles HTTP parsing, routing, and thread pool management. Because the KVStore is already thread-safe via its internal `shared_mutex`, the multi-threaded Crow server can handle concurrent requests without additional synchronization at the API layer.

10.2 The Building Process

The API exposes five endpoints:

- GET `/api/kv/<key>` — retrieve a value.
- POST `/api/kv` — insert or update a key (JSON body: `{"key": "...", "value": "..."}.`
- DELETE `/api/kv/<key>` — delete a key (idempotent tombstone write).
- GET `/api/metrics` — live engine telemetry (MemTable size, WAL size, SSTable count).
- GET `/api/memtable/snapshot` — full sorted dump of the active MemTable, consumed by the React SkipList visualizer.

The React dashboard uses two custom hooks: `useKVStore.js` for CRUD operations and `useMetrics.js` for polling `/api/metrics` every 2 seconds to update the live charts.

10.3 Implementation Details

The DELETE macro collision on Windows is handled explicitly (`#undef DELETE`) before the delete route is defined, preventing the Windows SDK's DELETE macro from interfering with Crow's method-matching DSL.

The server entry point in `main.cpp` initializes the engine with a configuration struct, replays the WAL (recovering from any previous crash), and starts the HTTP server. The entire startup sequence takes under 100 ms on typical hardware.

10.4 Code & Configuration

```

1 void ApiServer::setup_routes() {
2
3     // GET /api/kv/<key>
4     CROW_ROUTE(app_, "/api/kv/<string>").methods(crow::HTTPMethod::GET)
5     (
6         [this](const std::string& key) {
7             auto val = store_>get(key);
8             if (!val) return crow::response(404, "Key not found");
9             return crow::response(200, *val);
10        });
11
12    // POST /api/kv -- body: {"key": "...", "value": "..."}
13    CROW_ROUTE(app_, "/api/kv").methods(crow::HTTPMethod::POST)(
14        [this](const crow::request& req) {
15            auto x = crow::json::load(req.body);
16            if (!x || !x.has("key") || !x.has("value"))
17                return crow::response(400, "Invalid JSON or missing
18                    fields");
19            store_>put(std::string(x["key"].s()), std::string(x["value"]
20                .s()));
21            return crow::response(200, "OK");
22        });
23
24    // DELETE /api/kv/<key>
25    // Note: #undef DELETE required on Windows to suppress the SDK
26    // macro.
27    #ifdef DELETE
28    #undef DELETE
29    #endif
30    CROW_ROUTE(app_, "/api/kv/<string>").methods(crow::HTTPMethod::
31        DELETE)(
32        [this](const std::string& key) {
33            store_>remove(key);
34            return crow::response(200, "OK"); // Always 200 (idempotent
35                )
36        });
37
38    // GET /api/metrics -- live engine telemetry
39    CROW_ROUTE(app_, "/api/metrics").methods(crow::HTTPMethod::GET)(
40        [this]() {
41            crow::json::wvalue metrics;
42            metrics["memtable_size_bytes"] = store_>memtable_size();
43            metrics["wal_size_bytes"] = store_>wal_size();
44            metrics["sstable_count"] = store_>sstable_count();
45            return crow::response(metrics);
46        });
47    }

```

```

41 // GET /api/memtable/snapshot -- sorted MemTable dump for
42 // visualizer
43 CROW_ROUTE(app_, "/api/memtable/snapshot").methods(crow::HTTPMethod
44 ::GET)(
45     [this]() {
46         auto records = store->memtable_snapshot();
47         crow::json::wvalue::list arr;
48         for (const auto& rec : records) {
49             crow::json::wvalue entry;
50             entry["key"] = rec.key;
51             entry["value"] = rec.value;
52             entry["type"] = (rec.type == RecordType::DELETE)
53                 ? "tombstone" : "put";
54             arr.push_back(std::move(entry));
55         }
56         crow::json::wvalue result;
57         result["entries"] = std::move(arr);
58         result["count"] = records.size();
59         return crow::response(result);
60     }

```

Listing 13: REST API Route Definitions: src/api_routes.cpp

```

1 int main(int argc, char* argv[]) {
2     try {
3         kvault::EngineConfig config;
4         std::string base_dir = "data";
5
6         for (int i = 1; i < argc; ++i) {
7             if (std::string(argv[i]) == "--data-dir" && i + 1 < argc)
8                 base_dir = argv[++i];
9         }
10
11         config.wal_directory = base_dir + "/wal";
12         config.sstable_directory = base_dir + "/sstables";
13         config.server_port = 8080;
14
15         // KVStore constructor: creates directories, initializes
16         // subsystems,
17         // and replays the WAL to recover from any previous crash.
18         auto store = std::make_shared<kvault::KVStore>(config);
19
20         std::cout << "KVVault initialized.\n"
21             << "Recovered WAL size: " << store->wal_size()
22             << " bytes\n"
23             << "Active SSTables: " << store->sstable_count()
24             << "\n";
25
26         kvault::ApiServer server(store, config.server_port);
27         std::cout << "HTTP API on port " << config.server_port << " --
28             Ctrl+C to stop.\n";
29         server.run(); // Blocks; Crow manages the thread pool
30             internally.
31
32     } catch (const std::exception& e) {
33         std::cerr << "Fatal error: " << e.what() << "\n";
34     }
35 }

```

```
29     return 1;
30 }
31 return 0;
32 }
```

Listing 14: Server Entry Point: src/main.cpp

```
1 import { useState, useEffect, useCallback } from 'react';
2
3 const API_BASE = 'http://localhost:8080';
4
5 export function useMetrics(pollIntervalMs = 2000) {
6   const [metrics, setMetrics] = useState({
7     memtable_size_bytes: 0,
8     wal_size_bytes:      0,
9     sstable_count:        0,
10  });
11   const [error, setError] = useState(null);
12
13   const fetchMetrics = useCallback(async () => {
14     try {
15       const res = await fetch(`${API_BASE}/api/metrics`);
16       if (!res.ok) throw new Error(`HTTP ${res.status}`);
17       const data = await res.json();
18       setMetrics(data);
19       setError(null);
20     } catch (err) {
21       setError(err.message);
22     }
23   }, []);
24
25   useEffect(() => {
26     fetchMetrics();
27     // Poll /api/metrics every pollIntervalMs to keep the dashboard
28     // live.
29     const id = setInterval(fetchMetrics, pollIntervalMs);
30     return () => clearInterval(id); // Cleanup on unmount.
31   }, [fetchMetrics, pollIntervalMs]);
32
33   return { metrics, error };
```

Listing 15: React Metrics Hook: dashboard/src/hooks/useMetrics.js